In Kubernetes, **ReplicationController (RC)** and **ReplicaSet (RS)** both ensure that a specified number of pod replicas are running, but **ReplicaSet is the modern, recommended option**.

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1. High-level Difference

Aspect	**ReplicationController (RC)**	**ReplicaSet (RS)**
Status	**Deprecated / Legacy**	**Active & Recommended**
Introduced in	Early Kubernetes versions	Kubernetes 1.2+
Selector support	Equality-based selectors only	**Set-based selectors (more powerful)**
Used by Deployment	❌ No	✅ Yes
Rolling updates	Manual	**Handled by Deployment**
Scaling	Manual	Via Deployment / kubectl
Future support	Being phased out	Long-term supported

2. Selector Capability (Key Difference)

ReplicationController – Equality-based only

```
selector:
  app: nginx
```

ReplicaSet – Set-based (more flexible)

```
selector:
  matchExpressions:
  - key: app
    operator: In
    values:
    - nginx
    - web
```

➡️ **ReplicaSet can manage multiple labels and complex conditions**, RC cannot.

3. API Version

Resource	**API Version**
ReplicationController	v1
ReplicaSet	apps/v1

4. Relationship with Deployment (Very Important for Interviews)

- **ReplicationController**
  - ❌ Cannot be managed by Deployment
  - ❌ No rollout history
  - ❌ No rollback support
- **ReplicaSet**
  - ✅ Automatically created & managed by Deployment
  - ✅ Supports rolling updates
  - ✅ Supports rollback
  - ✅ Zero-downtime deployments

👉 **In real projects, you almost never create ReplicaSets directly**.
👉 You create a **Deployment**, and Kubernetes manages the ReplicaSet for you.

5. Example Manifests

ReplicationController (Legacy – Avoid)

```
apiVersion: v1
kind: ReplicationController
metadata:
 name: nginx-rc
spec:
 replicas: 3
 selector:
  app: nginx
 template:
  metadata:
   labels:
    app: nginx
  spec:
   containers:
   - name: nginx
    image: nginx
```

ReplicaSet (Modern)

```
apiVersion: apps/v1
kind: ReplicaSet
metadata:
 name: nginx-rs
spec:
 replicas: 3
 selector:
  matchLabels:
   app: nginx
 template:
  metadata:
   labels:
    app: nginx
  spec:
   containers:
   - name: nginx
    image: nginx
```

6. When to Use What?

Scenario	**Recommended**
New application	❌ RC / ❌ RS → ✅ **Deployment**
Legacy cluster maintenance	RC (only if already present)
Understanding internals	RS
Production workloads	**Deployment (manages RS)**

ReplicationController is a legacy controller with limited selector support, while ReplicaSet is its modern replacement with powerful set-based selectors and is used by Deployments for rolling updates and scaling.